

Matéria: Banco de Dados

Lista de Colaboradores

Felipe Durante
Guilherme da Silva Machado
Lucas Baldin Silva
Mateus da Silva Oliveira
Michael Moisés Alves de Moura
Pedro Henrique de Oliveira
Reverton Gonçalves
Samuel Silva Xavier
Vitória da Cunha Araújo
Vitor da Silva Santos

Sprint 3.2_TechMinds.pdf

Contexto

Vocês foram contratados como Analistas de Dados da TechMinds, uma startup em rápido

crescimento. O primeiro desafio de vocês é estruturar o banco de dados do RH para gerenciar os colaboradores, seus departamentos e salários, além de extrair relatórios estratégicos para a diretoria.

Atividade 1: Crie o banco de dados que abrigará as informações da empresa e ative-o para uso.

CREATE DATABASE rh_techminds; USE rh_techminds;

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Schema SQL

```
1 CREATE DATABASE rh_techminds;  
2 USE rh_techminds;  
3
```

Text to DDL

Query SQL

```
1
```

Results

There are no results to be displayed.

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https://fiddle.mysql.com/v5.7/run?source=db-fiddle&utm_medium=promo-ba...

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Atividade 2: Crie uma tabela chamada colaboradores. Ela deve ter um identificador único automático, o nome do funcionário (até 100 caracteres), o email (até 100 caracteres), o departamento (até 50 caracteres), o salário (número inteiro) e a data de admissão.

CREATE TABLE colaboradores (

id SERIAL PRIMARY KEY,

VARCHAR(100) NOT NULL, email

VARCHAR(100) UNIQUE,

departamento VARCHAR(50),

salario INT,

data_admissao DATE

);

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Schema SQL

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1 CREATE DATABASE rh_techminds;
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3 CREATE TABLE colaboradores (
4   id SERIAL PRIMARY KEY,
5   nome VARCHAR(100) NOT NULL,
6   email VARCHAR(100) UNIQUE,
7   departamento VARCHAR(50),
8   salario INT,
9   data_admissao DATE
10 );
11

```

Text to DDL

Query SQL

```

1

```

Results

There are no results to be displayed.

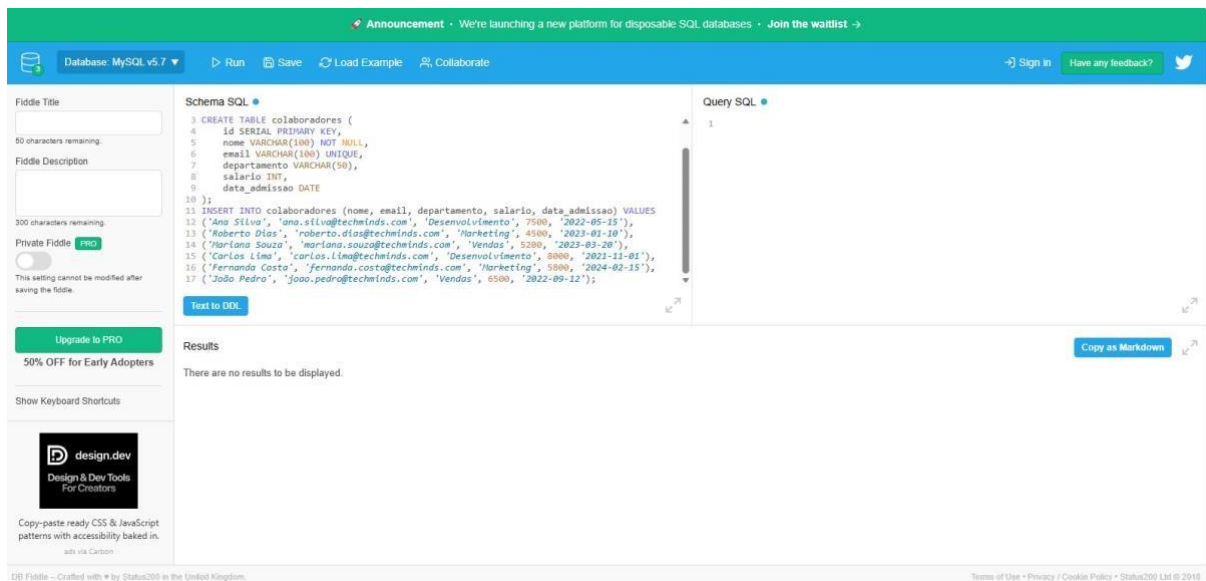
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Atividade 3: Insira 6 colaboradores na tabela recém-criada, distribuindo-os entre os departamentos de 'Desenvolvimento', 'Marketing' e 'Vendas'. Lembre-se das regras de aspas para textos e datas.

INSERT INTO colaboradores (nome, email, departamento, salario, data_admissao)
VALUES

('Ana Silva', 'ana.silva@techminds.com', 'Desenvolvimento', 7500, '2022-05-15'),
('Roberto Dias', 'roberto.dias@techminds.com', 'Marketing', 4500, '2023-01-10'),
('Mariana Souza', 'mariana.souza@techminds.com', 'Vendas', 5200, '2023-03-20'),
('Carlos Lima', 'carlos.lima@techminds.com', 'Desenvolvimento', 8000, '2021-11-01'),
('Fernanda Costa', 'fernanda.costa@techminds.com', 'Marketing', 5800, '2024-02-15'),
('João Pedro', 'joao.pedro@techminds.com', 'Vendas', 6500, '2022-09-12');



Atividade 4: O RH precisa de uma lista apenas com o nome e o email dos colaboradores que ganham mais de 6000. Escreva a consulta.

Query #1

```
SELECT nome, email
FROM colaboradores
WHERE salario > 6000;
```

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Schema SQL

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2 USE rh_techminds;
3 CREATE TABLE colaboradores (
4   id SERIAL PRIMARY KEY,
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15 ('Carlos Lima', 'carlos.lima@techminds.com', 'Desenvolvimento', 8000, '2021-11-01');
```

Text to DDL

Query SQL

```
1 SELECT nome, email
2 FROM colaboradores
3 WHERE salario > 6000;
```

Results

Copy as Markdown

Query #1 Execution time: 0.27ms

nome	email
Ana Silva	ana.silva@techminds.com
Carlos Lima	carlos.lima@techminds.com
João Pedro	joao.pedro@techminds.com

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Atividade 5: O colaborador 'Roberto Dias' foi promovido e seu salário passou a ser 6000. Atualize este dado no banco com segurança.

Query #2

```
UPDATE colaboradores
SET salario = 6000
WHERE nome = 'Roberto Dias';
```

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Schema SQL

```

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2 USE rh_techminds;
3 CREATE TABLE colaboradores (
4   id SERIAL PRIMARY KEY,
5   nome VARCHAR(100) NOT NULL,
6   email VARCHAR(100) UNIQUE,
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15 ('Carlos Lima', 'carlos.lima@techminds.com', 'Desenvolvimento', 8000, '2021-11-01');

```

Text to DDL

Query SQL

```

1 SELECT nome, email
2 FROM colaboradores
3 WHERE salario > 6000;
4 UPDATE colaboradores
5 SET salario = 6000
6 WHERE nome = 'Roberto Dias';

```

Results

Copy as Markdown

Query #1 Execution time: 0.21ms

nome	email
Ana Silva	ana.silva@techminds.com
Carlos Lima	carlos.lima@techminds.com
João Pedro	joao.pedro@techminds.com

Query #2 Execution time: 0.26ms

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Atividade 6: A colaboradora 'Mariana Souza' deixou a empresa. Remova o registro dela do banco de dados utilizando

DELETE FROM colaboradores

WHERE id = 3;

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Schema SQL

```

3 CREATE TABLE colaboradores (
4   id SERIAL PRIMARY KEY,
5   nome VARCHAR(100) NOT NULL,
6   email VARCHAR(100) UNIQUE,
7   departamento VARCHAR(50),
8   salario INT,
9   data_admissao DATE
10 );
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17 ('João Pedro', 'joao.pedro@techminds.com', 'Vendas', 6500, '2022-09-12');
```

[Text to DDL](#)

Query SQL

```

1 SELECT nome, email
2 FROM colaboradores
3 WHERE salario > 6000;
4 UPDATE colaboradores
5 SET salario = 6000
6 WHERE nome = 'Roberto Dias';
7 DELETE FROM colaboradores
8 WHERE id = 3;
```

Results [Copy as Markdown](#)

Carlos Lima	carlos.lima@techminds.com
João Pedro	joao.pedro@techminds.com

Query #2 **Execution time: 0.26ms**
There are no results to be displayed.

Query #3 **Execution time: 0.19ms**
There are no results to be displayed.

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Atividade 7: A diretoria quer saber os números totais da empresa: a quantidade total de funcionários ativos e a média salarial da startup.

SELECT COUNT(*) AS total_funcionarios, AVG(salario) AS media_salarial

FROM colaboradores;

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Schema SQL

```

1 CREATE DATABASE rh_techminds;
2 USE rh_techminds;
3 CREATE TABLE colaboradores (
4   id SERIAL PRIMARY KEY,
5   nome VARCHAR(100) NOT NULL,
6   email VARCHAR(100) UNIQUE,
7   departamento VARCHAR(50),
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15 ('Carlos Lima', 'carlos.lima@techminds.com', 'Desenvolvimento', 8000, '2021-11-01');

```

Text to DDL

Query SQL

```

1 SELECT nome, email
2 FROM colaboradores
3 WHERE salario > 6000;
4 UPDATE colaboradores
5 SET salario = 6000
6 WHERE nome = 'Roberto Dias';
7 DELETE FROM colaboradores
8 WHERE id = 3;
9 SELECT COUNT(*) AS total_funcionarios, AVG(salario) AS media_salarial
10 FROM colaboradores;

```

Copy as Markdown

Results

There are no results to be displayed.

Query #3 Execution time: 0.24ms

There are no results to be displayed.

Query #4 Execution time: 0.15ms

total_funcionarios	media_salarial
5	6760.0

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Atividade 8: Em vez do total da empresa, descubra quantos colaboradores existem em cada departamento separadamente.

SELECT departamento, COUNT(*) AS quantidade
FROM colaboradores
GROUP BY departamento;

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Schema SQL

```

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2 USE rh_techminds;
3 CREATE TABLE colaboradores (
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6   email VARCHAR(100) UNIQUE,
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8   salario INT,
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10 );
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```

Text to DDL

Query SQL

```

1 SELECT nome, email
2 FROM colaboradores
3 WHERE salario > 6000;
4 UPDATE colaboradores
5 SET salario = 6000
6 WHERE nome = 'Roberto Dias';
7 DELETE FROM colaboradores
8 WHERE id = 3;
9 SELECT COUNT(*) AS total_funcionarios, AVG(salario) AS media_salarial
10 FROM colaboradores;
11 SELECT departamento, COUNT(*) AS quantidade
12 FROM colaboradores
13 GROUP BY departamento;

```

Results

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Query #5 Execution time: 0.2ms

departamento	quantidade
Desenvolvimento	2
Marketing	2
Vendas	1

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Atividade 9: Para fins de auditoria, liste o maior salário (MAX) e o menor salário (MIN) pago em cada departamento.

**SELECT departamento, MAX(salario) AS maior_salario,
MIN(salario) AS menor_salario**

FROM colaboradores GROUP BY departamento;

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Schema SQL

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```

Text to DDL

Query SQL

```

2 FROM colaboradores
3 WHERE salario > 6000;
4 UPDATE colaboradores
5 SET salario = 6000
6 WHERE nome = 'Roberto Dias';
7 DELETE FROM colaboradores
8 WHERE id = 3;
9 SELECT COUNT(*) AS total_funcionarios, AVG(salario) AS media_salarial
10 FROM colaboradores;
11 SELECT departamento, COUNT(*) AS quantidade
12 FROM colaboradores
13 GROUP BY departamento;
14 SELECT departamento, MAX(salario) AS maior_salario, MIN(salario) AS menor_salario
15 FROM colaboradores
16 GROUP BY departamento;

```

Results

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Query #6 Execution time: 0.12ms

departamento	maior_salario	menor_salario
Desenvolvimento	8000	7500
Marketing	6000	5800
Vendas	6500	6500

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Atividade 10: Escreva uma consulta avançada que aplique todos os filtros. A diretoria quer ver apenas os departamentos que possuem um custo total com salários (SUM) maior que 10000, mas considerando para o cálculo apenas os funcionários contratados em 2023 ou antes.

SELECT departamento, SUM(salario) AS custo_total

FROM colaboradores

WHERE data_admissao <= '2023-12-31' -- Contratados em 2023 ou antes

GROUP BY departamento

HAVING SUM(salario) > 10000; -- Apenas departamentos que custam mais de 10k



Database: MySQL v5.7 ▼



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Schema SQL

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```

[Text to DDL](#)

Query SQL

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2 FROM colaboradores
3 WHERE salario > 6000;
4 UPDATE colaboradores
5 SET salario = 6000
6 WHERE nome = 'Roberto Dias';
7 DELETE FROM colaboradores
8 WHERE id = 3;
9 SELECT COUNT(*) AS total_funcionarios, AVG(salario) AS media_salarial
10 FROM colaboradores;
11 SELECT departamento, COUNT(*) AS quantidade
12 FROM colaboradores
13 GROUP BY departamento;
14 SELECT departamento, MAX(salario) AS maior_salario, MIN(salario) AS menor_salario
15 FROM colaboradores
```

Results

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Marketing	6000	5800
Vendas	6500	6500

Query #7 Execution time: 0.13ms

departamento	custo_total
Desenvolvimento	15500

